

GED Math Practice Set: Equations, Inequalities, and Functions

November 6, 2025

Instructions: Show your work. Calculators are allowed unless otherwise noted. Round decimals to the nearest hundredth when needed.

Practice Questions (20)

1. Solve for x : $2x + 7 = 25$.
2. Solve for x : $5(x - 3) - 2 = 3x + 4$.
3. Solve for x : $\frac{x}{3} - \frac{5}{6} = \frac{1}{2}$.
4. Solve for x : $4(2x - 1) = 3(x + 5)$.
5. Solve for x : $7 - 2x = 3x + 12$.
6. A cell phone plan charges a monthly fee of \$20 plus \$0.10 per text message. If one month the bill was \$41, how many texts were sent?
7. Solve the inequality and state the solution set in interval notation: $3x - 4 \leq 11$.
8. Solve the compound inequality: $-2 \leq 2x - 6 < 8$.
9. Solve for x : $|x - 5| = 9$.
10. Solve the inequality: $|2x + 1| < 7$.
11. Solve the system by substitution: $\begin{cases} y = 2x - 3 \\ x + y = 12 \end{cases}$.
12. Solve the system by elimination: $\begin{cases} 3x + 2y = 16 \\ 5x - 2y = 4 \end{cases}$.
13. Two types of tickets were sold for a school event: student tickets for \$6 and adult tickets for \$10. If 180 tickets were sold for a total of \$1,520, how many of each type were sold?
14. Let $f(x) = -3x + 7$. Find $f(-2)$ and $f\left(\frac{1}{3}\right)$.
15. Determine whether the relation $\{(2, 3), (4, 5), (2, 6), (7, 8)\}$ is a function. Explain.
16. Find the slope-intercept form of the line through $(-3, 2)$ and $(5, -6)$.

17. Solve by factoring: $x^2 - 9x + 20 = 0$.
18. Solve using the quadratic formula: $2x^2 + 3x - 5 = 0$.
19. For $y = x^2 - 6x + 11$, find the axis of symmetry and the vertex.
20. Let $f(x) = 2x + 1$ and $g(x) = x^2 - 4$. Compute $(f \circ g)(3)$ and $(g \circ f)(3)$.

Full Solutions

1. $2x + 7 = 25 \Rightarrow 2x = 18 \Rightarrow x = 9$.
2. $5(x - 3) - 2 = 3x + 4$
 $5x - 15 - 2 = 3x + 4$
 $5x - 17 = 3x + 4$
 $2x = 21 \Rightarrow x = \frac{21}{2} = 10.5$.
3. $\frac{x}{3} - \frac{5}{6} = \frac{1}{2}$
Add $\frac{5}{6}$ to both sides: $\frac{x}{3} = \frac{1}{2} + \frac{5}{6} = \frac{3}{6} + \frac{5}{6} = \frac{8}{6} = \frac{4}{3}$.
Multiply both sides by 3: $x = 4$.
4. $4(2x - 1) = 3(x + 5)$
 $8x - 4 = 3x + 15$
 $5x = 19 \Rightarrow x = \frac{19}{5} = 3.8$.
5. $7 - 2x = 3x + 12$
 $7 - 12 = 3x + 2x$
 $-5 = 5x \Rightarrow x = -1$.
6. Let t be the number of texts. Cost: $20 + 0.10t = 41$.
 $0.10t = 21 \Rightarrow t = 210$.
Answer: 210 texts.
7. $3x - 4 \leq 11$
 $3x \leq 15 \Rightarrow x \leq 5$.
Interval: $(-\infty, 5]$.
8. $-2 \leq 2x - 6 < 8$
Add 6: $4 \leq 2x < 14$
Divide by 2: $2 \leq x < 7$.
Interval: $[2, 7)$.
9. $|x - 5| = 9$
 $x - 5 = 9$ or $x - 5 = -9 \Rightarrow x = 14$ or $x = -4$.
10. $|2x + 1| < 7$
 $-7 < 2x + 1 < 7$
 $-8 < 2x < 6$
 $-4 < x < 3$.
Interval: $(-4, 3)$.
11. Substitute $y = 2x - 3$ into $x + y = 12$:
 $x + (2x - 3) = 12 \Rightarrow 3x = 15 \Rightarrow x = 5$.
Then $y = 2(5) - 3 = 7$.
Solution: $(5, 7)$.

12. Add the equations to eliminate y :

$$(3x + 2y) + (5x - 2y) = 16 + 4 \text{ gives } 8x = 20 \Rightarrow x = \frac{20}{8} = \frac{5}{2}.$$

Plug into $3x + 2y = 16$: $3 \cdot \frac{5}{2} + 2y = 16$

$$\frac{15}{2} + 2y = 16 \Rightarrow 2y = 16 - \frac{15}{2} = \frac{17}{2}$$

$$y = \frac{17}{4} = 4.25.$$

Solution: $\left(\frac{5}{2}, \frac{17}{4}\right)$.

13. Let s be student tickets, a be adult tickets.

System:
$$\begin{cases} s + a = 180 \\ 6s + 10a = 1520 \end{cases}$$

From first, $s = 180 - a$. Substitute: $6(180 - a) + 10a = 1520$

$$1080 - 6a + 10a = 1520 \Rightarrow 4a = 440 \Rightarrow a = 110.$$

Then $s = 180 - 110 = 70$.

Answer: 70 student, 110 adult.

14. $f(-2) = -3(-2) + 7 = 6 + 7 = 13$.

$$f\left(\frac{1}{3}\right) = -3 \cdot \frac{1}{3} + 7 = -1 + 7 = 6.$$

15. Not a function. The input $x = 2$ maps to two different outputs (3 and 6), which violates the definition of a function (each input must have exactly one output).

16. Slope $m = \frac{-6 - 2}{5 - (-3)} = \frac{-8}{8} = -1$.

Use point-slope with $(-3, 2)$: $y - 2 = -1(x + 3)$.

$$y - 2 = -x - 3 \Rightarrow y = -x - 1.$$

Slope-intercept form: $y = -x - 1$.

17. Factor: $x^2 - 9x + 20 = (x - 4)(x - 5) = 0$.

Solutions: $x = 4$ or $x = 5$.

18. Quadratic formula with $a = 2, b = 3, c = -5$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-3 \pm \sqrt{9 - 4(2)(-5)}}{4} = \frac{-3 \pm \sqrt{9 + 40}}{4} = \frac{-3 \pm \sqrt{49}}{4}.$$

Thus $x = \frac{-3 + 7}{4} = 1$ or $x = \frac{-3 - 7}{4} = -\frac{10}{4} = -\frac{5}{2}$.

19. Complete the square or use $x = -\frac{b}{2a}$. Here $a = 1, b = -6$.

Axis: $x = -\frac{-6}{2(1)} = \frac{6}{2} = 3$.

Vertex: Evaluate y at $x = 3$: $y = 3^2 - 6(3) + 11 = 9 - 18 + 11 = 2$.

Axis: $x = 3$. **Vertex:** $(3, 2)$.

20. $(f \circ g)(3) = f(g(3))$.

$$g(3) = 3^2 - 4 = 9 - 4 = 5. \text{ Then } f(5) = 2(5) + 1 = 11.$$

$$(g \circ f)(3) = g(f(3)). \text{ } f(3) = 2(3) + 1 = 7. \text{ Then } g(7) = 7^2 - 4 = 49 - 4 = 45.$$

Answers: $(f \circ g)(3) = 11, (g \circ f)(3) = 45$.